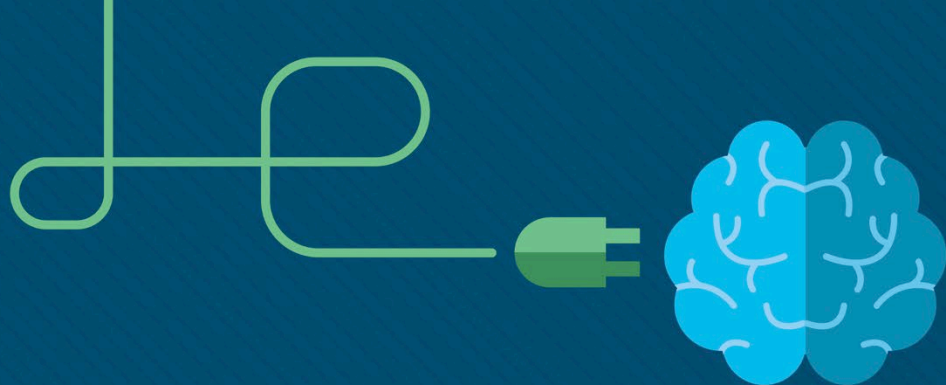




Module 16: Troubleshoot Static and Default Routes

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: Troubleshoot Static and Default Routes

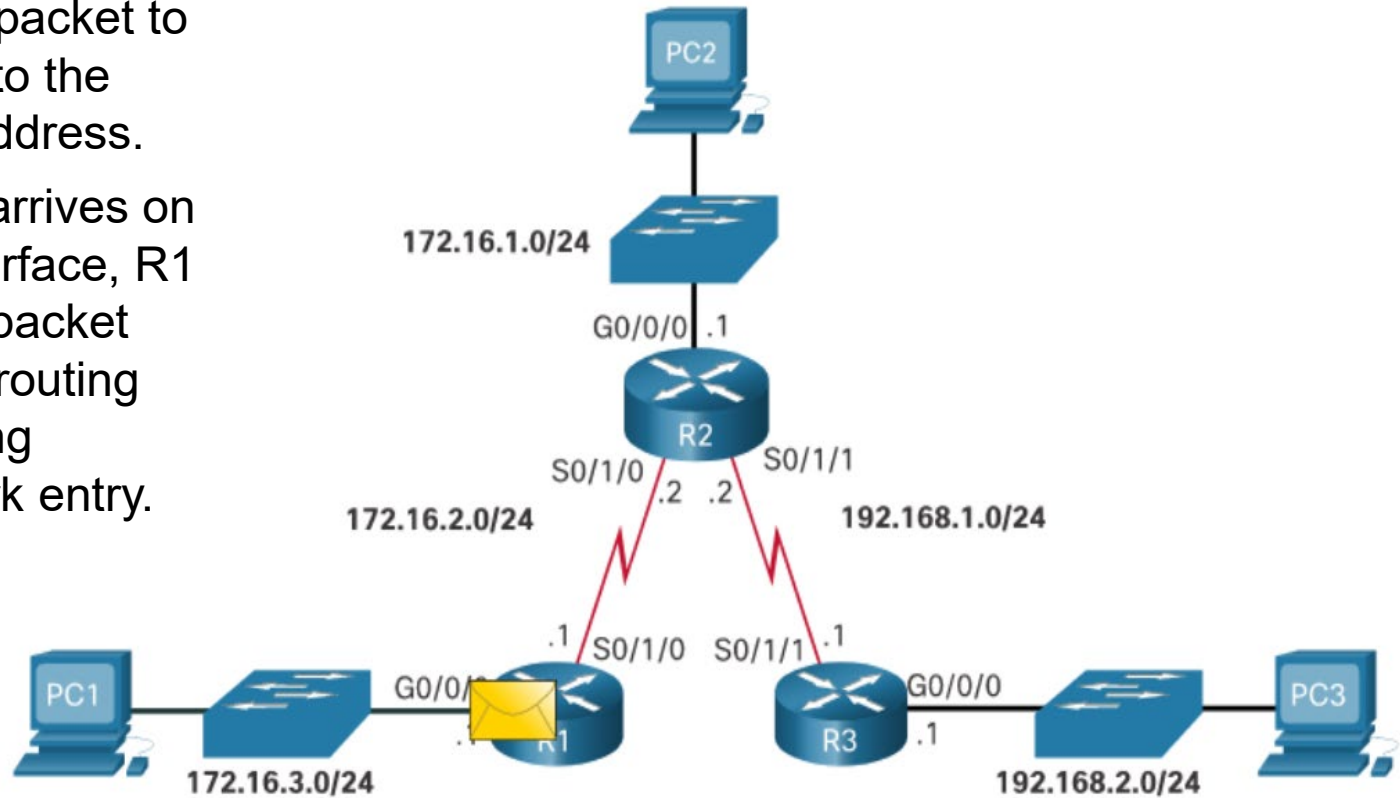
Module Objective: Troubleshoot static and default route configurations.

Topic Title	Topic Objective
Packet Processing with Static Routes	Explain how a router processes packets when a static route is configured.
Troubleshoot IPv4 Static and Default Route Configuration	Troubleshoot common static and default route configuration issues.

16.1 Packet Processing with Static Routes

Static Routes and Packet Forwarding

- PC1 addresses a packet to PC3 and sends it to the default gateway address.
- When the packet arrives on the R1 G0/0/0 interface, R1 decapsulates the packet and searches the routing table for a matching destination network entry.

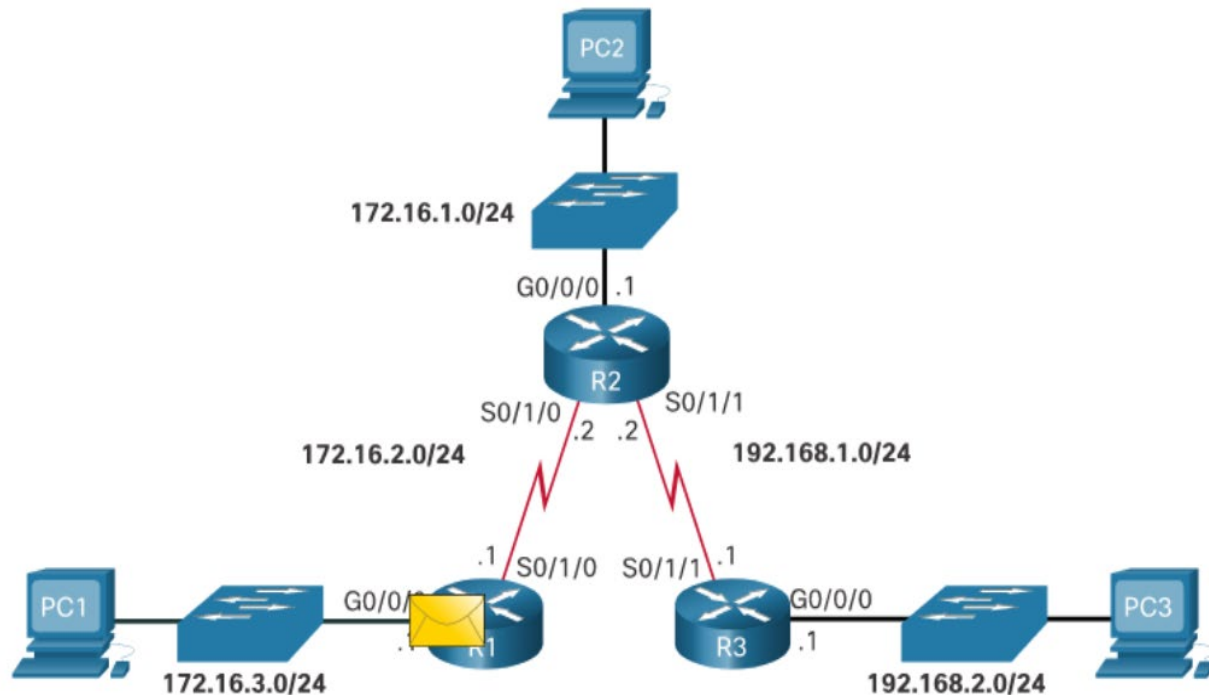


Packet Processing with Static Routes

Static Routes and Packet Forwarding

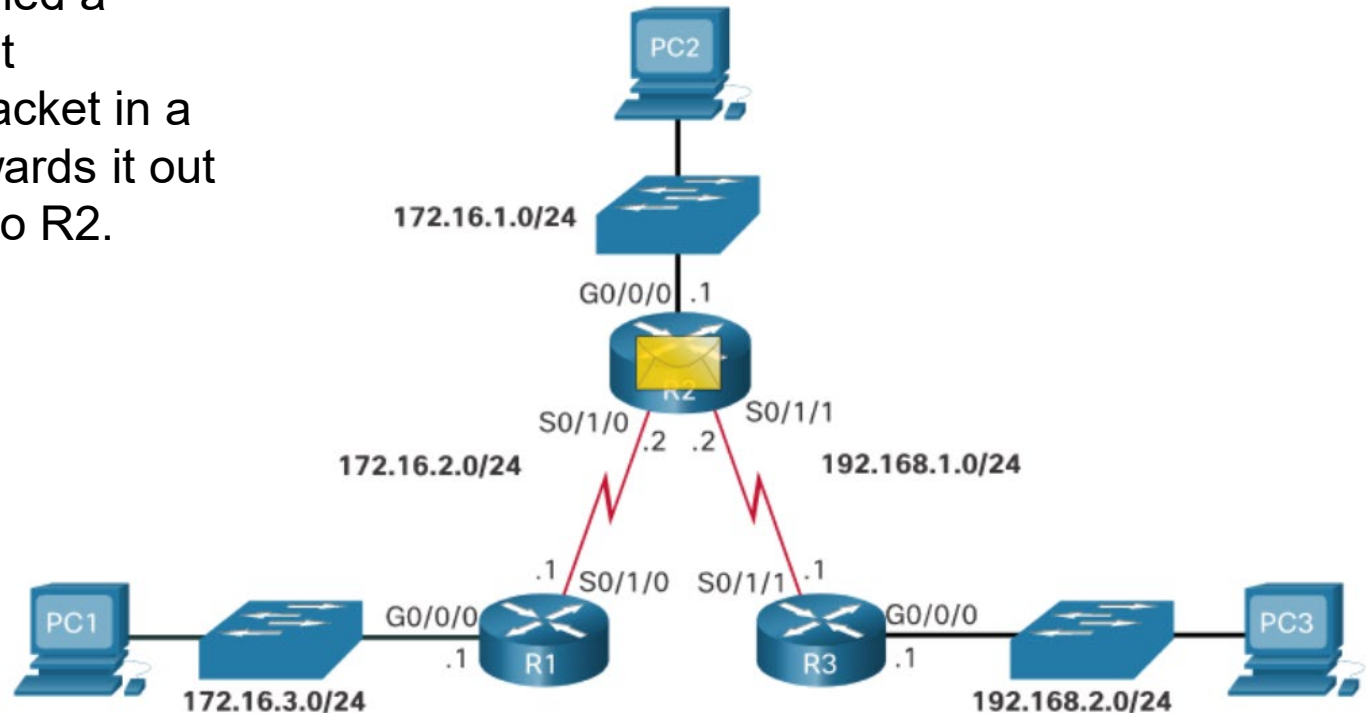
If the destination IP address:

- Matches a static route entry, R1 will use the static route to identify the next-hop IP address or exit interface.
- Does not match a specific route to the destination network, then R1 will use the default static route (if configured).
- Does not match a route table entry, then R1 will drop the packet and send an ICMP message back to the source (i.e., PC1).



Static Routes and Packet Forwarding (Cont.)

Assuming R1 matched a routing table entry, it encapsulates the packet in a new frame and forwards it out of interface S0/1/0 to R2.

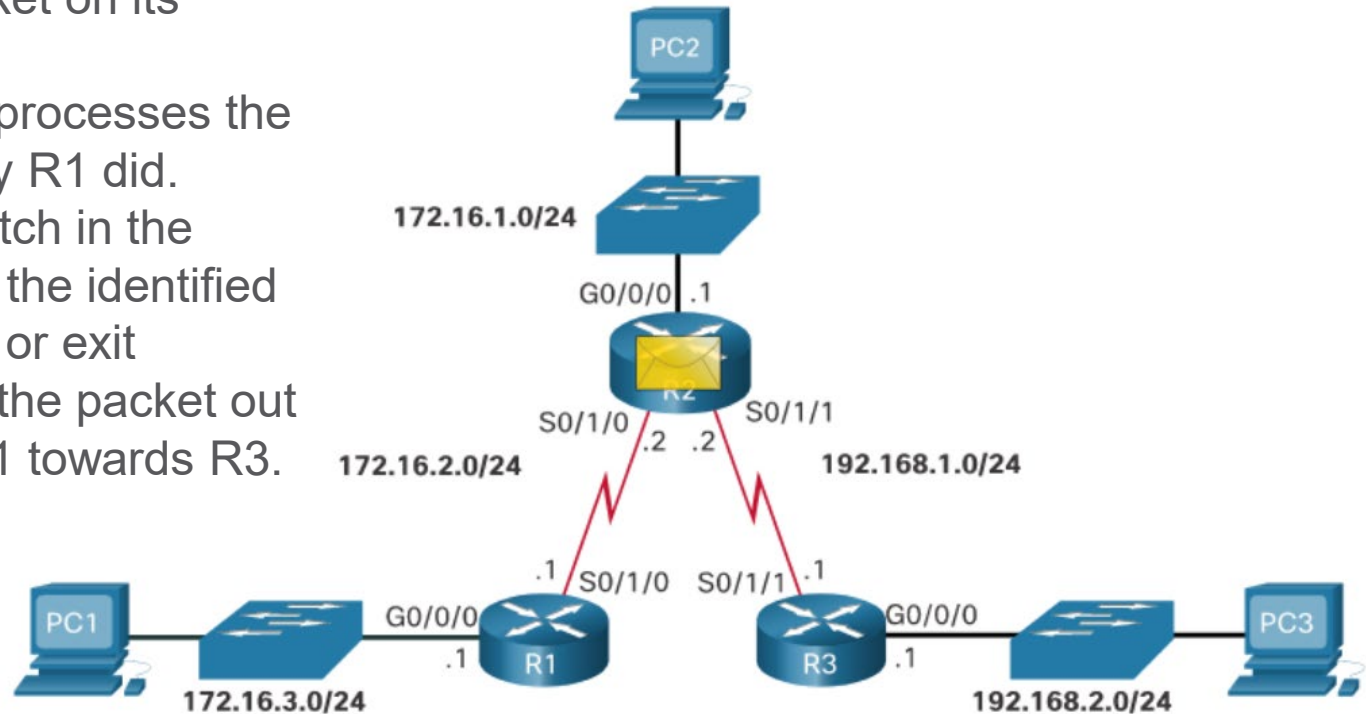


Static Routes and Packet Forwarding (Cont.)

R2 receives the packet on its S0/1/0 interface.

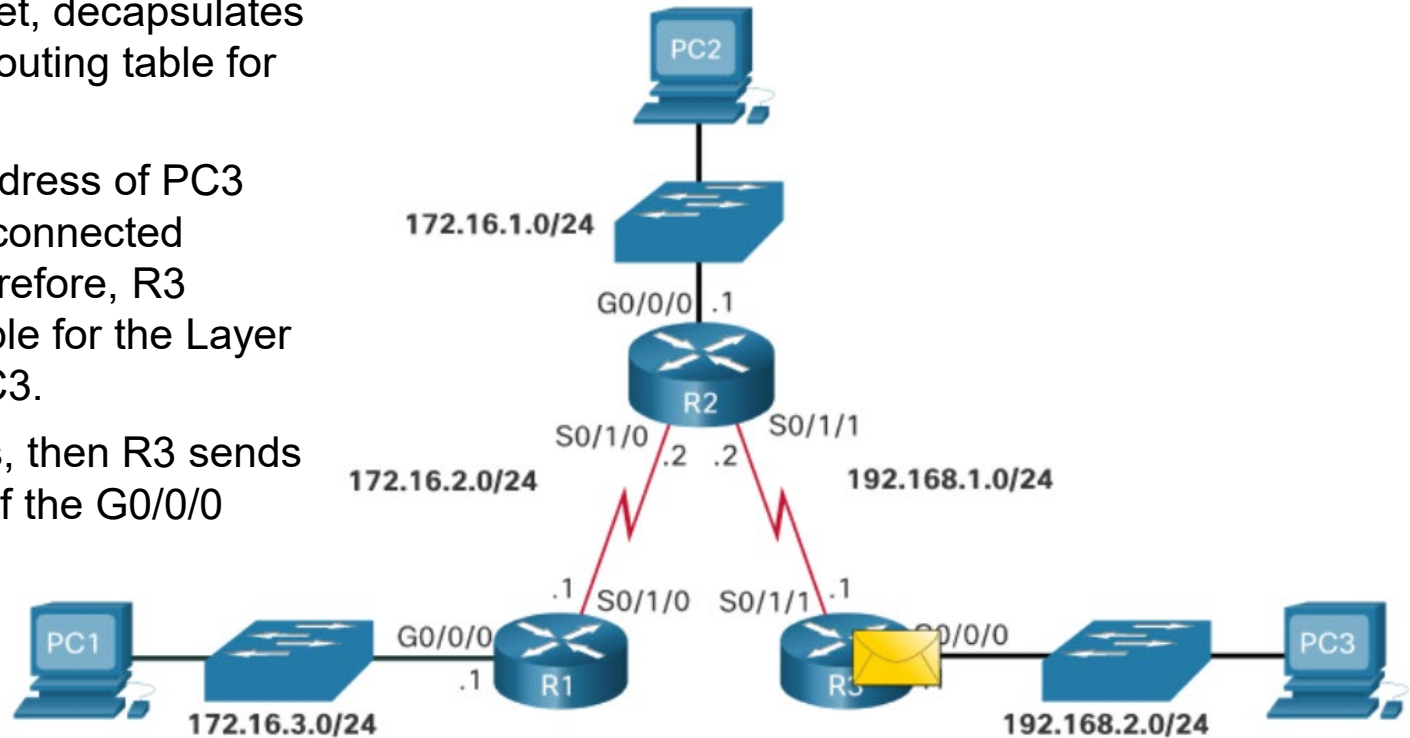
It decapsulates and processes the packet the same way R1 did.

When R2 finds a match in the routing table, it uses the identified next-hop IP address or exit interface and sends the packet out of its interface S0/1/1 towards R3.



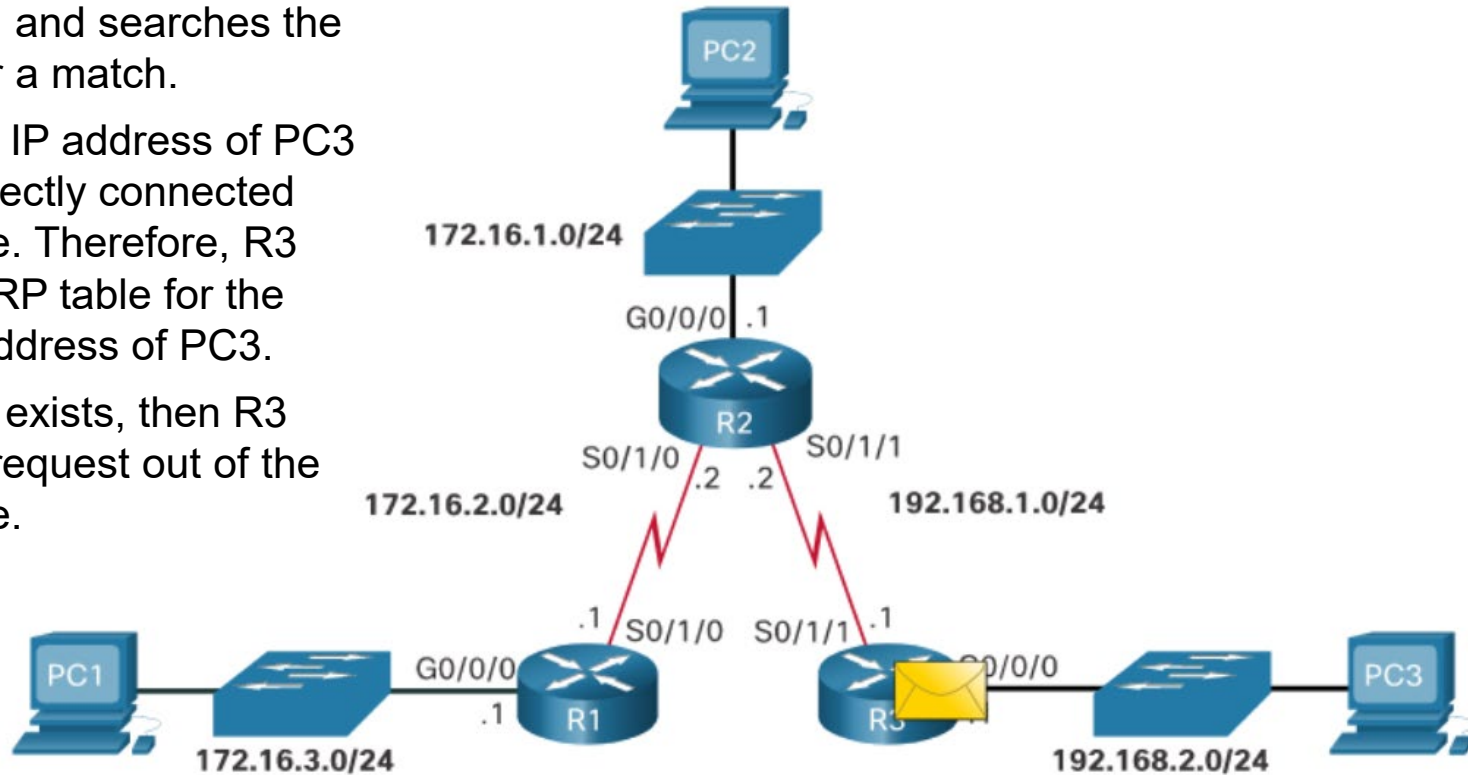
Static Routes and Packet Forwarding (Cont.)

- R3 receives the packet, decapsulates it, and searches the routing table for a match.
- The destination IP address of PC3 matches the directly connected G0/0/0 interface. Therefore, R3 searches the ARP table for the Layer 2 MAC address of PC3.
- If no ARP entry exists, then R3 sends an ARP request out of the G0/0/0 interface.



Static Routes and Packet Forwarding (Cont.)

- R3 receives the packet, decapsulates it, and searches the routing table for a match.
- The destination IP address of PC3 matches the directly connected G0/0/0 interface. Therefore, R3 searches the ARP table for the Layer 2 MAC address of PC3.
- If no ARP entry exists, then R3 sends an ARP request out of the G0/0/0 interface.



Static Routes and Packet Forwarding (Cont.)

- PC3 responds with an ARP reply containing its MAC address.
- R3 encapsulates the packet in a new frame and uses the PC3 MAC address as the destination MAC address and the G0/0/0 MAC address as the source MAC address.
- The frame is forwarded out of interface G0/0/0 and PC3 receives and processes it accordingly.

